

Python Programming Fundamentals Cheat Sheet

| Package/Method   | Description                                                                                         | Syntax and Code Example                                                                                                                                                                                                                                                              |
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| AND              | Returns `True` if both statement1 and statement2 are `True`. Otherwise, returns `False`.            | <p>Syntax:</p> <pre>statement1 and statement2</pre> <p>Example:</p> <pre>marks = 90 attendance_percentage = 87 if marks &gt;= 80 and attendance_percentage &gt;= 85:     print("qualify for honors") else:     print("Not qualified for honors") # Output = qualify for honors</pre> |
| Class Definition | Defines a blueprint for creating objects and defining their attributes and behaviors.               | <p>Syntax:</p> <pre>class ClassName: # Class attributes and methods</pre> <p>Example:</p> <pre>class Person:     def __init__(self, name, age):         self.name = name         self.age = age</pre>                                                                                |
| Define Function  | A `function` is a reusable block of code that performs a specific task or set of tasks when called. | <p>Syntax:</p> <pre>def function_name(parameters): # Function body</pre> <p>Example:</p> <pre>def greet(name): print("Hello,", name)</pre>                                                                                                                                           |

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| Equal(==) | Checks if two values are equal.                                                                                                                   | <p>Syntax:</p> <pre>variable1 == variable2</pre> <p>Example 1:</p> <pre>5 == 5</pre> <p>returns True</p> <p>Example 2:</p> <pre>age = 25 age == 30</pre> <p>returns False</p>                                                                                        |
| For Loop  | A 'for' loop repeatedly executes a block of code for a specified number of iterations or over a sequence of elements (list, range, string, etc.). | <p>Syntax:</p> <pre>for variable in sequence: # Code to repeat</pre> <p>Example 1:</p> <pre>for num in range(1, 10):     print(num)</pre> <p>Example 2:</p> <pre>fruits = ["apple", "banana", "orange", "grape", "kiwi"] for fruit in fruits:     print(fruit)</pre> |

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| Function Call                | A function call is the act of executing the code within the function using the provided arguments. | <p>Syntax:</p> <pre>function_name(arguments)</pre> <p>Example:</p> <pre>greet("Alice")</pre>                                                                                                                                      |
| Greater Than or Equal To(>=) | Checks if the value of variable1 is greater than or equal to variable2.                            | <p>Syntax:</p> <pre>variable1 &gt;= variable2</pre> <p>Example 1:</p> <pre>5 &gt;= 5 and 9 &gt;= 5</pre> <p>returns True</p> <p>Example 2:</p> <pre>quantity = 105 minimum = 100 quantity &gt;= minimum</pre> <p>returns True</p> |
| Greater Than(>)              | Checks if the value of variable1 is greater than variable2.                                        | <p>Syntax:</p> <pre>variable1 &gt; variable2</pre>                                                                                                                                                                                |

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|              |                                                                                                                                                       | <p>Example 1: <math>9 &gt; 6</math></p> <p>returns True</p> <p>Example 2:</p> <pre>age = 20 max_age = 25 age &gt; max_age</pre> <p>returns False</p>                                                                                                                                                                                                                                        |
| If Statement | Executes code block `if` the condition is `True`.                                                                                                     | <p>Syntax:</p> <pre>if condition: #code block for if statement</pre> <p>Example:</p> <pre>if temperature &gt; 30:     print("It's a hot day!")</pre>                                                                                                                                                                                                                                        |
| If-Elif-Else | Executes the first code block if condition1 is `True`, otherwise checks condition2, and so on. If no condition is `True`, the else block is executed. | <p>Syntax:</p> <pre>if condition1:     # Code if condition1 is True elif condition2:     # Code if condition2 is True else:     # Code if no condition is True</pre> <p>Example:</p> <pre>score = 85 # Example score if score &gt;= 90:     print("You got an A!") elif score &gt;= 80:     print("You got a B.") else:     print("You need to work harder.") # Output = You got a B.</pre> |

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| If-Else Statement         | Executes the first code block if the condition is `True`, otherwise the second block. | <p>Syntax:</p> <pre>if condition: # Code, if condition is True else: # Code, if condition is False</pre> <p>Example:</p> <pre>if age &gt;= 18:     print("You're an adult.") else:     print("You're not an adult yet.")</pre> |
| Less Than or Equal To(<=) | Checks if the value of variable1 is less than or equal to variable2.                  | <p>Syntax:</p> <pre>variable1 &lt;= variable2</pre> <p>Example 1:</p> <pre>5 &lt;= 5 and 3 &lt;= 5</pre> <p>returns True</p> <p>Example 2:</p> <pre>size = 38 max_size = 40 size &lt;= max_size</pre> <p>returns True</p>      |

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| Less Than(<)  | Checks if the value of variable1 is less than variable2.                                                                | <p>Syntax:</p> <pre>variable1 &lt; variable2</pre> <p>Example 1:</p> <pre>4 &lt; 6</pre> <p>returns True</p> <p>Example 2:</p> <pre>score = 60 passing_score = 65 score &lt; passing_score</pre> <p>returns True</p>                                                                                                                                                |
| Loop Controls | 'break' exits the loop prematurely. 'continue' skips the rest of the current iteration and moves to the next iteration. | <p>Syntax:</p> <pre>for: # Code to repeat     if # boolean statement         break for: # Code to repeat     if # boolean statement         continue</pre> <p>Example 1:</p> <pre>for num in range(1, 6):     if num == 3:         break     print(num)</pre> <p>Example 2:</p> <pre>for num in range(1, 6):     if num == 3:         continue     print(num)</pre> |

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| NOT           | Returns `True` if variable is `False`, and vice versa. | <p>Syntax:</p> <pre>not variable</pre> <p>Example:</p> <pre>isLocked = False<br/>print(not isLocked)</pre> <p>returns True if the variable is False (i.e., unlocked).</p>                             |
| Not Equal(!=) | Checks if two values are not equal.                    | <p>Syntax:</p> <pre>variable1 != variable2</pre> <p>Example:</p> <pre>a = 10<br/>b = 20<br/>a != b</pre> <p>returns True</p> <p>Example 2:</p> <pre>count=0<br/>count != 0</pre> <p>returns False</p> |

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| Object Creation | Creates an instance of a class (object) using the class constructor.                                | <p>Syntax:</p> <pre>object_name = ClassName(arguments)</pre> <p>Example:</p> <pre>person1 = Person("Alice", 25)</pre>                                                                                                                                                                 |
| OR              | Returns `True` if either statement1 or statement2 (or both) are `True`. Otherwise, returns `False`. | <p>Syntax:</p> <pre>statement1 or statement2</pre> <p>Example:</p> <pre>"Farewell Party Invitation" grade = 12 if grade == 11 or grade == 12:     print("Farewell Party Invitation") else:     print("Not eligible")</pre> <p>returns True</p>                                        |
| range()         | Generates a sequence of numbers within a specified range.                                           | <p>Syntax:</p> <pre>range(stop) range(start, stop) range(start, stop, step)</pre> <p>Example:</p> <pre>range(5) #generates a sequence of integers from 0 to 4. range(2, 10) #generates a sequence of integers from 2 to 9. range(1, 11, 2) #generates odd integers from 1 to 9.</pre> |



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| Return Statement           | 'Return' is a keyword used to send a value back from a function to its caller.                                                      | <p>Syntax:</p> <pre>return value</pre> <p>Example:</p> <pre>def add(a, b): return a + b result = add(3, 5)</pre>                                                                                                                                                                                                                                                 |
| Try-Except Block           | Tries to execute the code in the try block. If an exception of the specified type occurs, the code in the except block is executed. | <p>Syntax:</p> <pre>try: # Code that might raise an exception except ExceptionType: # Code to handle the exception</pre> <p>Example:</p> <pre>try:     num = int(input("Enter a number: ")) except ValueError:     print("Invalid input. Please enter a valid number.")</pre>                                                                                    |
| Try-Except with Else Block | Code in the 'else' block is executed if no exception occurs in the try block.                                                       | <p>Syntax:</p> <pre>try: # Code that might raise an exception except ExceptionType: # Code to handle the exception else: # Code to execute if no exception occurs</pre> <p>Example:</p> <pre>try:     num = int(input("Enter a number: ")) except ValueError:     print("Invalid input. Please enter a valid number") else:     print("You entered:", num)</pre> |

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| Try-Except with Finally Block | Code in the 'finally' block always executes, regardless of whether an exception occurred.           | <p>Syntax:</p> <pre>try: # Code that might raise an exception except ExceptionType: # Code to handle the exception finally: # Code that always executes</pre> <p>Example:</p> <pre>try:     file = open("data.txt", "r")     data = file.read() except FileNotFoundError:     print("File not found.") finally:     file.close()</pre> |
| While Loop                    | A 'while' loop repeatedly executes a block of code as long as a specified condition remains 'True'. | <p>Syntax:</p> <pre>while condition: # Code to repeat</pre> <p>Example:</p> <pre>count = 0 while count &lt; 5:     print(count)     count += 1</pre>                                                                                                                                                                                   |



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